Algebraic substitution

Combining expressions

Substitute an expression and collect like terms.

10 MINUTES / 4 QUESTIONS / AS-07

Name _____ Date _____

Show your working. Use brackets for negative values.

Starter: use the full set. Exit ticket: choose 2-3 questions.

1 Given $a = 2b + 2$

Write in terms of b and simplify

Simplify $5a + 4b$

2 Given $a = 3b + 5$

Write in terms of b and simplify

Simplify $2a + 5b$

3 Given $a = 6b + 1$

Write in terms of b and simplify

Simplify $2a + 6b$

4 Given $a = 2b - 5$

Write in terms of b and simplify

Simplify $3a + 6b$

Answer sheet

Combining expressions | 4 questions

Keep this section separate from the question sheet. Full working follows on the next pages.

01 $14b + 10$

02 $11b + 10$

03 $18b + 2$

04 $12b - 15$

Worked solutions

Combining expressions

01 Given $a = 2b + 2$

$5a + 4b$

Substitute $a = 2b + 2$.

$5(2b + 2) + 4b$

$10b + 10 + 4b$

$(10 + 4)b + 10 = 14b + 10$

02 Given $a = 3b + 5$

$2a + 5b$

Substitute $a = 3b + 5$.

$2(3b + 5) + 5b$

$6b + 10 + 5b$

$(6 + 5)b + 10 = 11b + 10$

03 Given $a = 6b + 1$

$2a + 6b$

Substitute $a = 6b + 1$.

$2(6b + 1) + 6b$

$12b + 2 + 6b$

$(12 + 6)b + 2 = 18b + 2$

04 Given $a = 2b - 5$

$3a + 6b$

Substitute $a = 2b - 5$.

$3(2b - 5) + 6b$

$6b - 15 + 6b$

$(6 + 6)b - 15 = 12b - 15$